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Appendix

A Datasets and Implementations

A.1 Selected Relations and Prompt Templates

We employed a set of criteria to select relations from Wikidata in order to construct our dataset. Our criteria largely follow the framework proposed by Gekhman et al. (2025). Specifically, we require that each factual query in the dataset be unambiguous: given a subject–relation pair, the object should be unique and easy verifiable. The criteria are as follows:

- **Avoid granularity ambiguity.** We exclude relations whose answers can vary in their level of detail. For example, in *location* queries, the response could be expressed as a city, state, or country, making it ill-defined (Yona et al., 2024).

Relation	Prompt Template
father	The name of the father of [subject] is
mother	The name of the mother of [subject] is
spouse	The name of the spouse of [subject] is
date of birth	The birth date of [subject] is

Table 3: Relations and prompt templates for querying factual knowledge of models. [subject] is a placeholder replaced with subject entities.

- **Avoid surface-level guessing.** We exclude relations whose correct answers can often be inferred from shallow patterns. For instance, *country of citizenship* can frequently be guessed from shallow lexical patterns, rather than reflecting actual memorization (Mallen et al., 2023b).

Following these criteria, Gekhman et al. (2025) narrowed the 24 relations introduced by Sciavolino et al. (2021) down to four. However, we observe that their filtering primarily addresses ambiguity at the relation and object levels, but does not consider ambiguity at the subject level. In practice, some relations involve subjects that are inherently ambiguous. For example, the relation *record label* can be problematic because many songs share identical names, leading to unclear subject–object mappings.

To mitigate such cases, we apply an additional subject-level filtering step and restrict our dataset to relations where the subject is a person, thereby reducing ambiguity. In addition, we manually include certain relations to strengthen the dataset. Concretely, we use the following four relations: P22 (*father*), P25 (*mother*), P26 (*spouse*), and P569 (*date of birth*). We show the list of the templates used to create our dataset in Table 3.

A.2 Implementation Details

Checkpoints and GPU resources. All the checkpoints used in our experiments are provided by the Hugging Face Transformers library (Wolf et al., 2019). Specifically, we use the checkpoint “meta-llama/Meta-Llama-3-8B”¹ and “mistralai/Mistral-7B-v0.3”² for the experiments of response generation (§3), hidden-state analysis (§4) and accessing the performance of hallucination detection methods (§5). For refusal tuning (§6), we use checkpoints

¹<https://huggingface.co/meta-llama/Meta-Llama-3-8B>

²<https://huggingface.co/mistralai/Mistral-7B-v0.3>

LLM Judge Prompt
I will give you a factual query (e.g., “The name of the father of <subj>”), a gold answer to the factual query, and a proposed answer generated by an LLM. You need to compare the proposed answer to the gold answer and assign it one of the possible grades using the steps below. Possible grades are: A: CORRECT B: INCORRECT C: WRONG GOLD D: ERROR Spelling errors, synonyms, abbreviations, or hedging expressions (e.g., “it is possible that”) should not alter the grade if the person referred to in the proposed answer matches the gold answer. Steps: Step 1: If the gold answer does not correspond to an answer for the question, output “C” and finish. Otherwise, proceed to Step 2. Step 2: Extract all predicted entities from the proposed answer. Proceed to Step 3. Step 3: If each predicted entity refers to the answer mentioned in the gold answer, output “A” and finish. Otherwise, proceed to Step 4. Step 4: If the predicted entity does not refer to the gold answer, output “B” and finish. Otherwise, proceed to Step 5. Step 5: Double-check whether the proposed answer refers to a different answer from the gold answer. If it does, output “B.” Otherwise, output “D” and finish. Input format: Question: {question} Gold answer: {gold_answer} Proposed answer: {proposed_answer} Instruction: Output your reasoning steps. After that, conclude your response with “Output:” followed by the letter (A, B, C, or D). Do not provide any further explanation.

Figure 11: LLM Judge prompt used for evaluation.

provided by the Unsloth framework (Daniel Han and team, 2023), namely “unsloth/llama-3-8b”³ and “unsloth/mistral-7b-v0.3”⁴, which enable more efficient fine-tuning. All experiments are conducted on 4 NVIDIA L40S GPUs.

Decoding algorithm. We employ greedy decoding (temperature = 0) for response generation, with models run in BF16 precision.

PEFT settings for refusal tuning. For refusal tuning, we fine-tune with both models using QLoRA (Dettmers et al., 2023), implemented with the Unsloth framework (Daniel Han and team, 2023), with rank $r = 8$, and $\alpha = 8$. QLoRA adapters are applied to all attention and MLP modules, and each model is fine-tuned for one epoch.

³<https://huggingface.co/unsloth/llama-3-8b>

⁴<https://huggingface.co/unsloth/mistral-7b-v0.3>

A.3 Labeling Factual Correctness

Here, we outline the process of determine Factual Correctness of a model prediction. We construct correctness labels through a two-stage process. First, we use spaCy⁵ Named Entity Recognizer to extract the target entity from the model’s output. If it matches the ground truth, the answer is marked correct. Otherwise, or if extraction fails, we rely on Qwen2.5-14B-Instruct (Yang et al., 2024a) as a judge to compare the predicted answer with the ground truth. Following Gekhman et al. (2025), we design the evaluation prompt, which is shown in Figure 11.

B Calculation of Subject Representation Norms

To investigate the geometric properties of subject representations, we compute the average L_2 norm

⁵<https://spacy.io/>

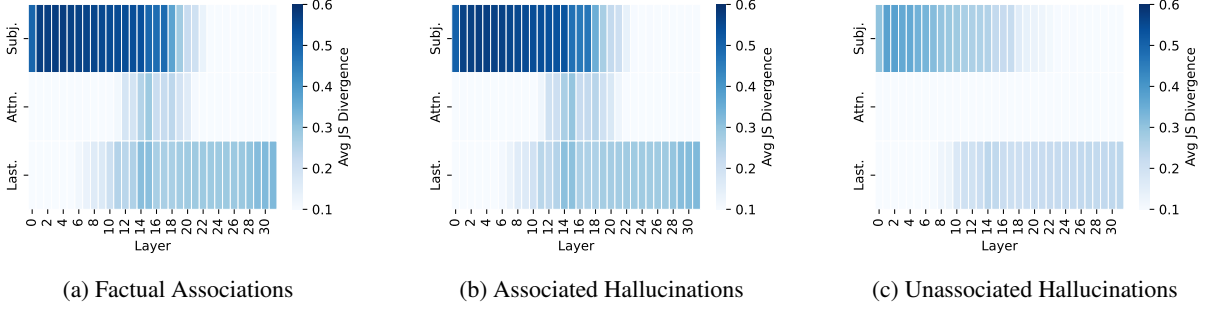


Figure 12: Effect of interventions across layers of Mistral-7B-v0.3. The heatmap shows JS divergence between the output distribution before and after intervention. Darker color indicates that the intervened hidden states are more causally influential on the model’s predictions. Top row: patching representations of subject tokens. Middle row: blocking attention flow from subject to the last token. Bottom row: patching representations of the last token.

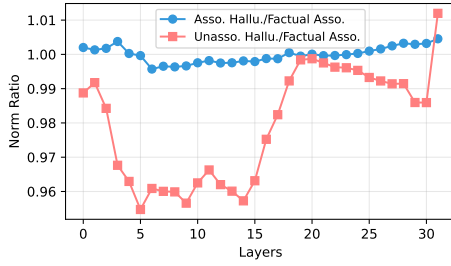


Figure 13: Norm ratio curves of subject representations in Mistral-7B-v0.3, comparing AHs and UHs against FAs as the baseline. At earlier layers, the norm of UH samples is significantly lower than that of AH samples.

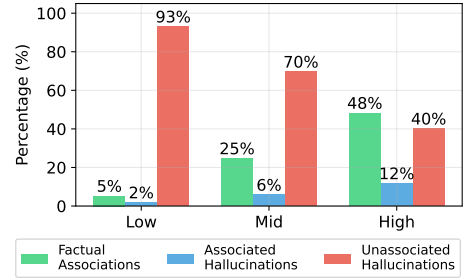


Figure 15: Sample distribution across different subject popularity (low, mid, high) in Mistral-7B-v0.3, measured by monthly Wikipedia page views.

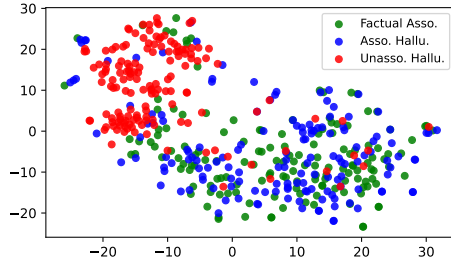


Figure 14: t-SNE visualization of last token’s representations at layer 25 of Mistral-7B-v0.3.

of the hidden states across different knowledge categories. For a single sample with subject tokens $s_1, \dots, s_n$, we first compute the average norm across all subject tokens at layer ℓ :

$$\bar{N}_{\text{sample}}^{(\ell)} = \frac{1}{n} \sum_{i=1}^n \left\| \mathbf{h}_{s_i}^{(\ell)} \right\|_2 \quad (2)$$

Then, for a given category (e.g., FAs, UHs, or AHs) containing K samples, we aggregate these values to obtain the per-category average norm:

$$\bar{N}_{\text{category}}^{(\ell)} = \frac{1}{K} \sum_{j=1}^K \bar{N}_{\text{sample},j}^{(\ell)} \quad (3)$$

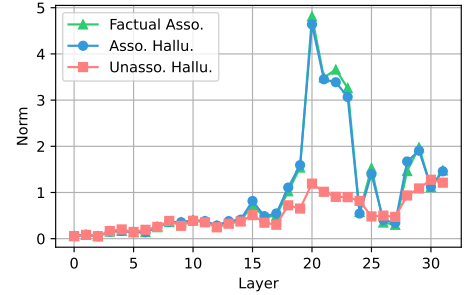


Figure 16: Subject-to-last attention contribution norms across layers in Mistral-7B-v0.3. Values show the norm of the attention contribution from subject tokens to the last token at each layer.

where $\bar{N}_{\text{sample},j}^{(\ell)}$ is the average norm for the j -th sample in the category. By repeating this process for each layer $\ell \in \{1, \dots, L\}$, we obtain a layer-wise norm profile. Finally, to facilitate direct comparison, we report the norm ratio of hallucinations relative to factual associations: